A Complexity-Theoretic Approach To Proofs of Space

Marshall Ball[1]

Jiaxin Guan[2]

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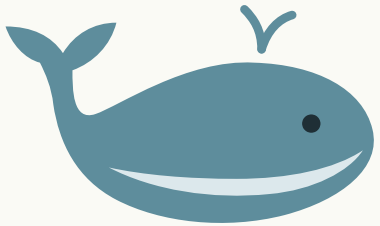
TCC 2026 · November 11, 2026

Proofs of Space
[Dziembowski, Faust, Kolmogorov, Pietrzak 15]

Introduced as an alternative to proof of work.

Useful for spam and Sybil resistance.

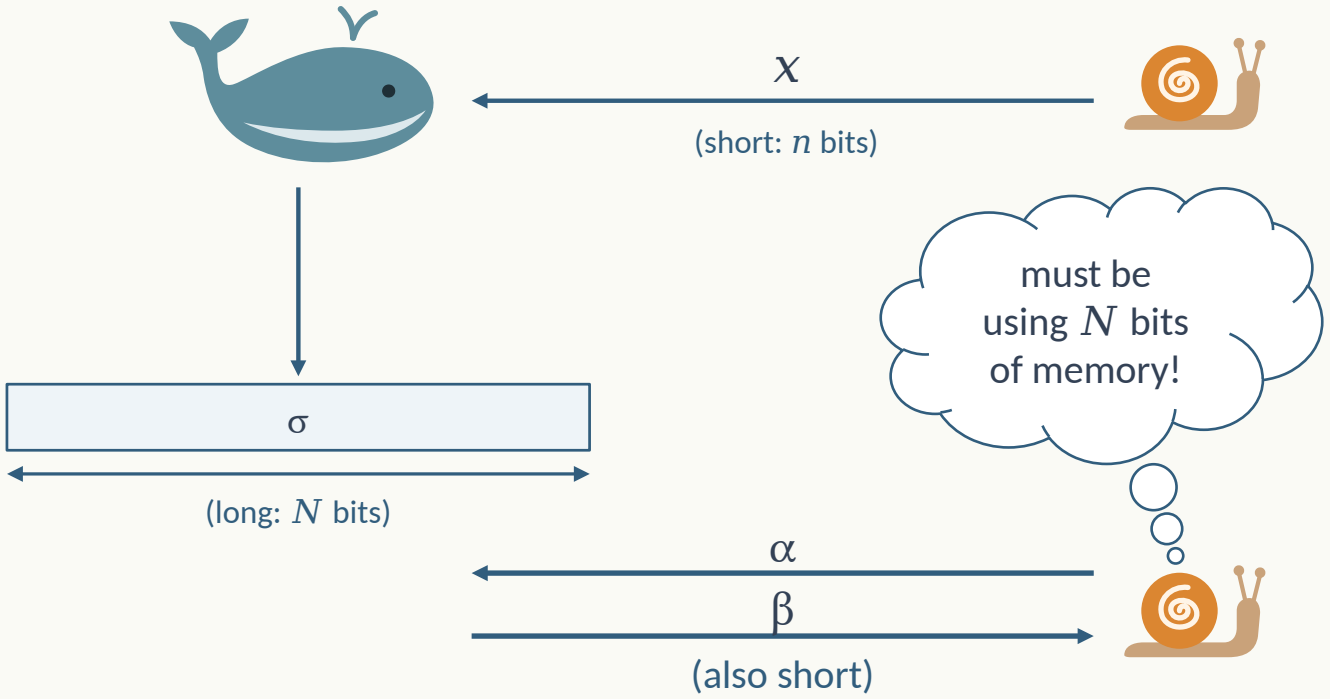
Connections to pebbling and memory-hard functions.



A **prover** convinces a tiny **verifier** that it persistently stores N bits.



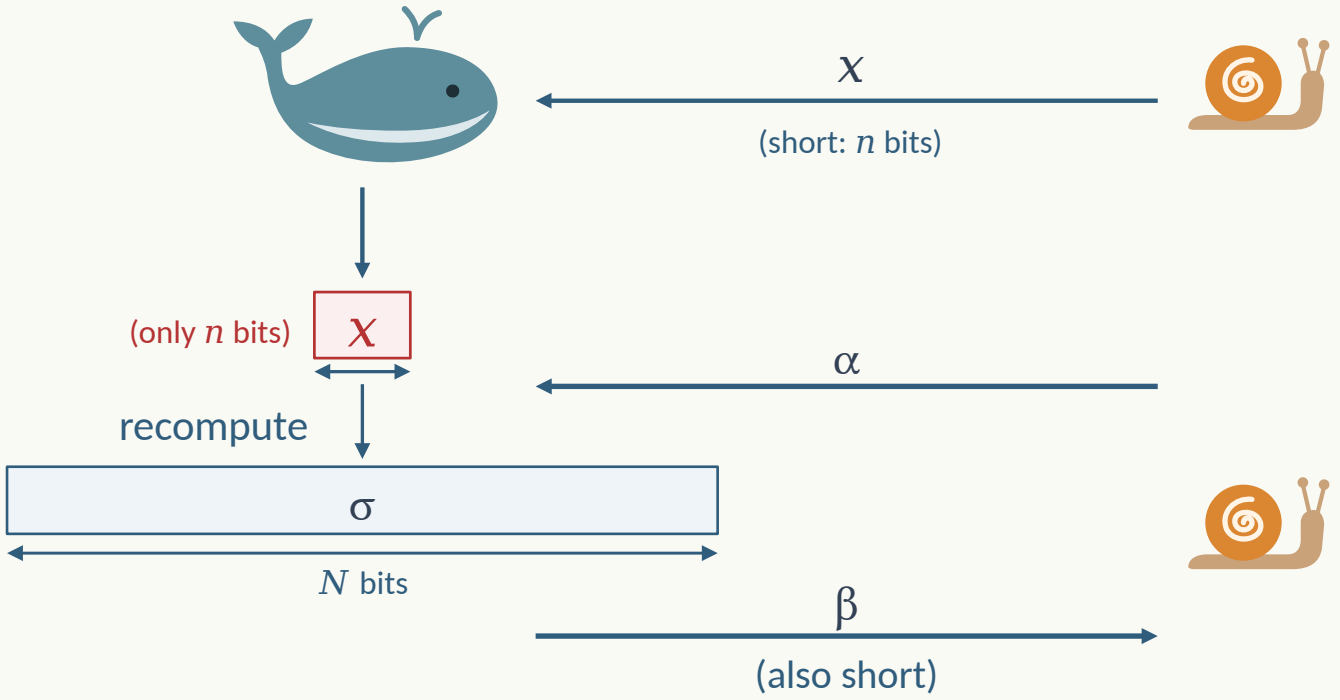
Proofs of Space [DFKP15]



Tiny verifier: All sent and received messages must be short.

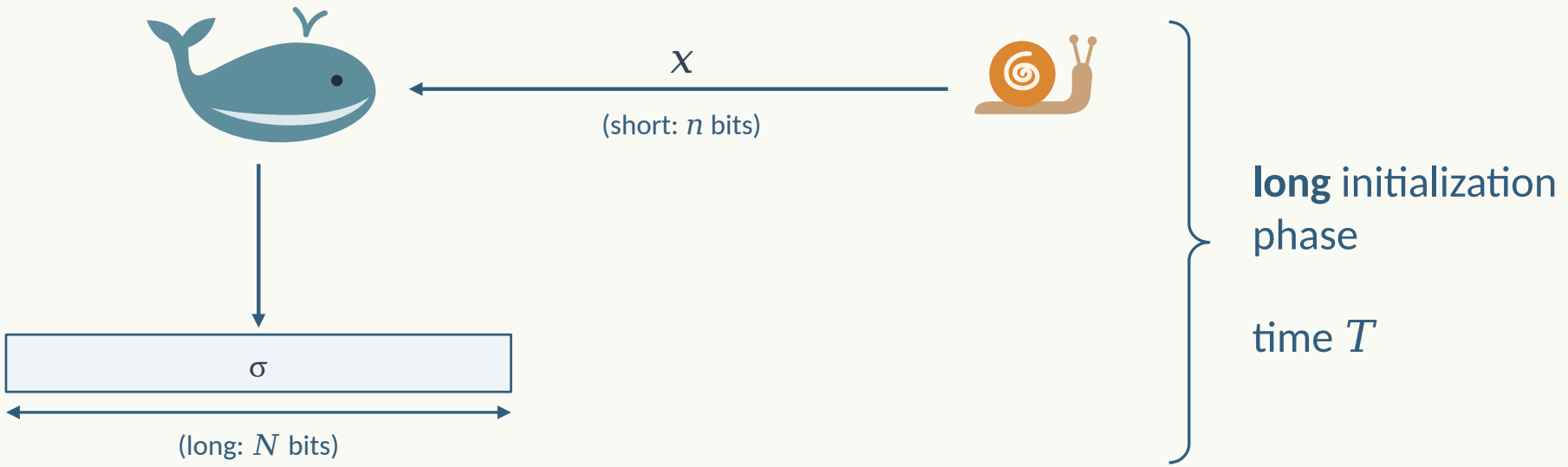
Otherwise, a large verifier could trivially send N random bits for the prover to store.

Why Not Just Store x?



Store only x; recompute σ when challenged.

Proofs of Space: Soundness



Sample Parameters

$n = \text{polylog}(N)$

$T = N^{100}$

$t = N^3$

Proof of Space Constructions

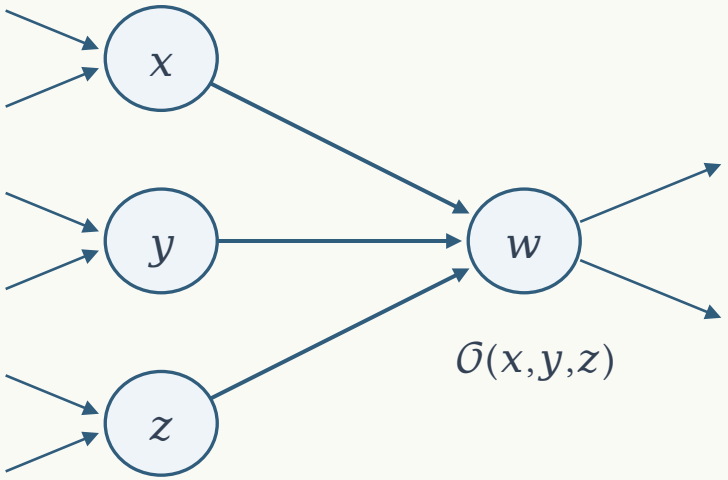
All previous constructions rely on

random oracles.

Their security proofs use both incompressibility and extractability.

How Everyone Else Builds PoS

Games on graphs



[DFKP15, RD16, ABP17, ABP18, Fis19, Rey24, …]

1. Incompressibility

Labels are truly random.

Compute and store the labels of sinks

2. Extractability (unpredictable)

The adversary is forced to interact with $\mathcal{O}$;
the reduction sees its queries.

Proof sketch: extract a good pebbling strategy from a too-good prover's queries to $\mathcal{O}$.

Open problem: formulate a natural assumption about cryptographic hashes that suffices.

This Work

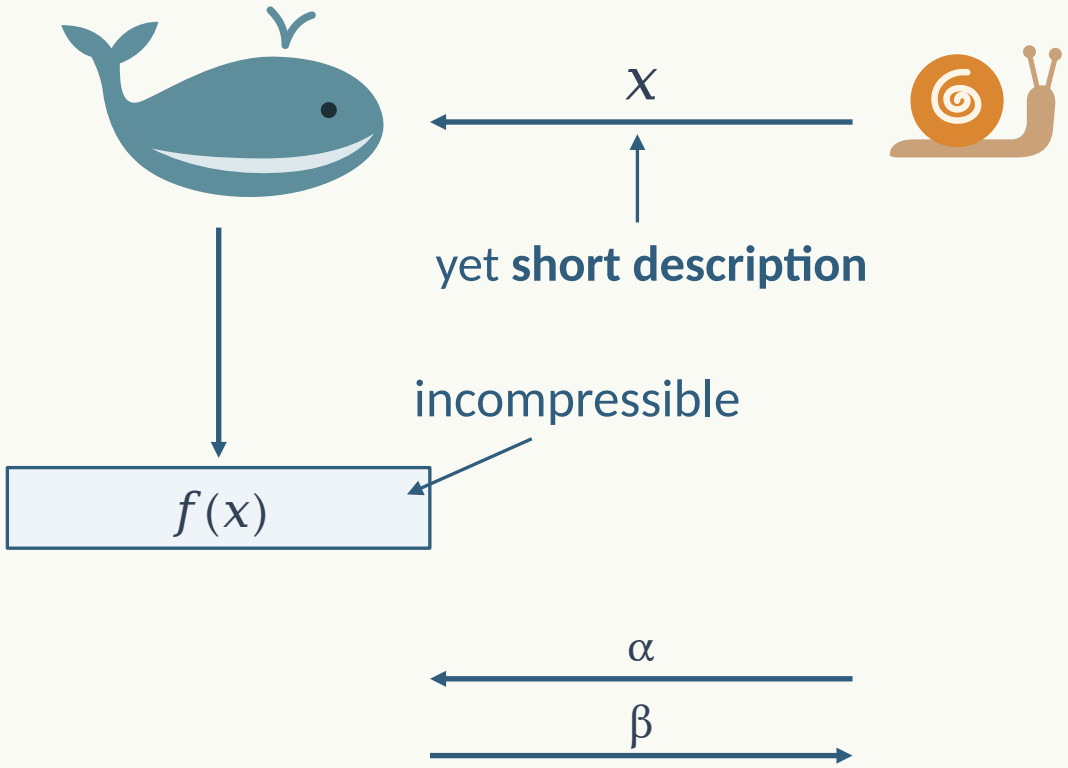
A simple, explicit framework

without random oracles

Our Contributions

- **Linking Proofs of Space to *computational depth***

A Contradiction?



need $f(x)$ such that:

① **incompressible** (must store σ)

w.r.t. time-t decompression

② **compressible** (short description x)

w.r.t. time-T decompression

$\Rightarrow$ $f(x)$ has high **"computational depth"**

[Antunes, Fortnow, van Melkebeek, Vinodchandran 06]

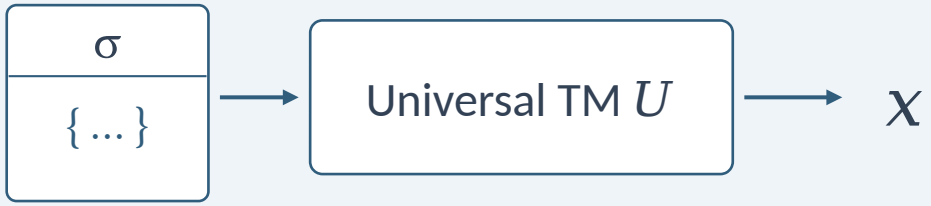
Computational Depth

[Antunes, Fortnow, van Melkebeek, Vinodchandran 06]

Time-bounded Kolmogorov complexity

$$K^t(x) := \min_\sigma \{ |\sigma| : \sigma \overset{t}{\rightsquigarrow} x \}$$

= length of shortest σ s.t. the universal TM U on input σ outputs x in t steps



$$K^t(x \mid y) := \min_\sigma \{ |\sigma| : (\sigma, y) \overset{t}{\rightsquigarrow} x \}$$

Computational depth

$$D^t(x) := K^t(x) - K^\infty(x)$$

measure of "nonrandom" / "useful" information in x

x is **deep** if it has a short explanation, but is hard to reconstruct from any short explanation (quickly).

Functions of High Computational Depth

$f : \{0,1\}^n \longrightarrow \{0,1\}^N$ has **high computational depth** if

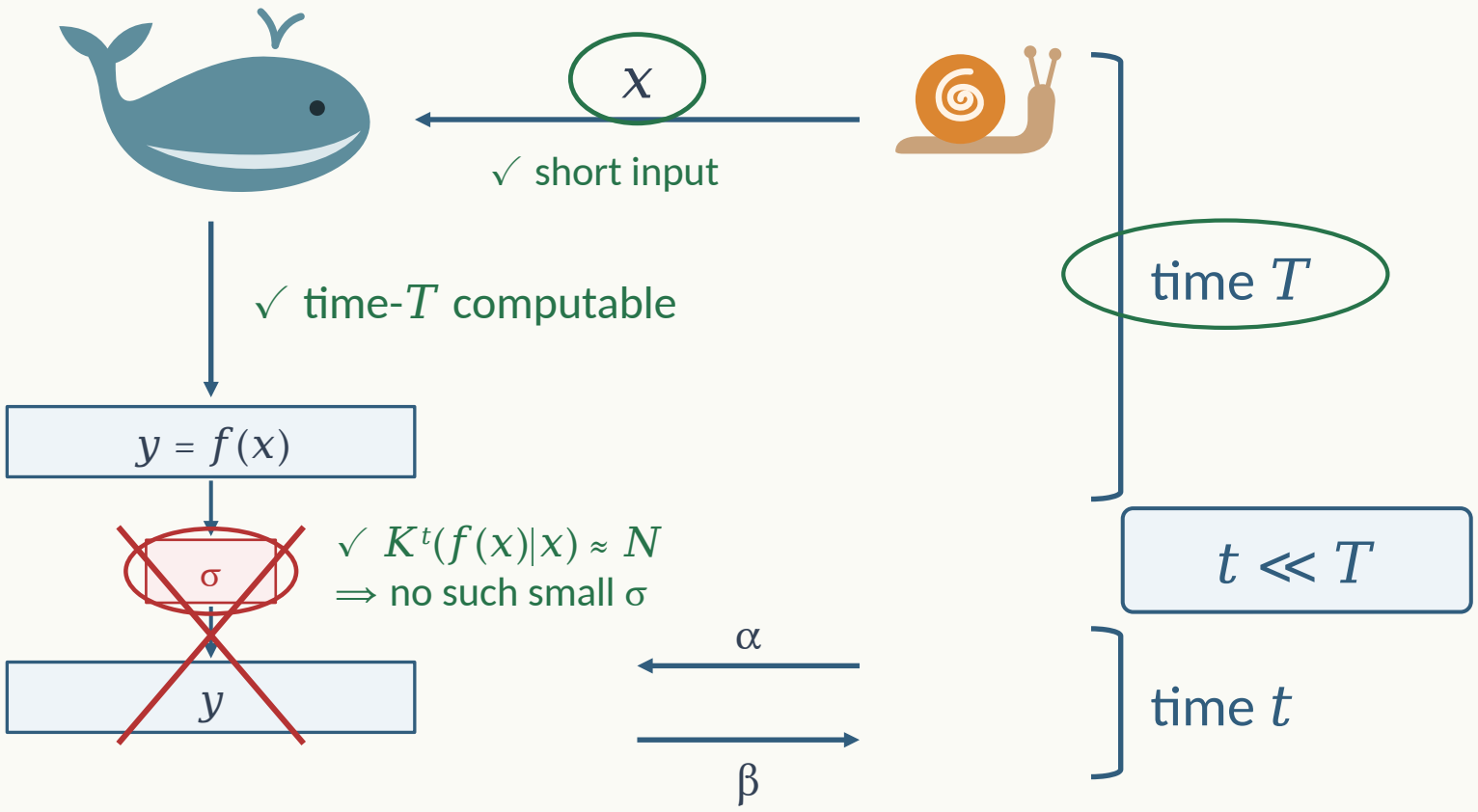
(i) input short $(n = \text{polylog}(N))$

(ii) f **computable in time** T

(iii) $K^t(f(x) \mid x) \approx N$ w.h.p. over x

($f(x)$ incompressible w.r.t. time t, even given x)

Match Made in Heaven



The definitions line up perfectly:

(i) Short input

short challenge x

(ii) Time-T computable

long initialization

(iii) Time-t incompressible

no small σ survives the short audit

Our Contributions

- Linking Proofs of Space to computational depth

- **Instantiating “deep” functions**

How Do We Get a "Deep" Function?

(at a very high level)

PRGs for nondeterministic distinguishers

[KvMS12, SU05]

A short seed generates a long string that looks random
even to nondeterministic time-t distinguishers.

Why does this give depth?

A short, fast description could be guessed and checked nondeterministically — distinguishing the output from random.

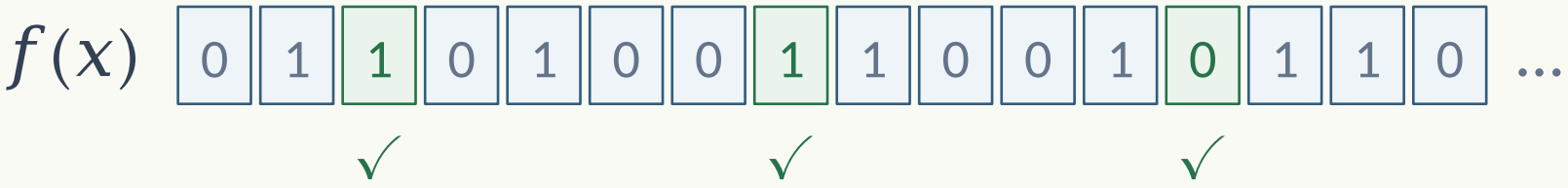
Our Contributions

- Linking Proofs of Space to computational depth
- Instantiating “deep” functions
- **An explicit Proof of Space framework**

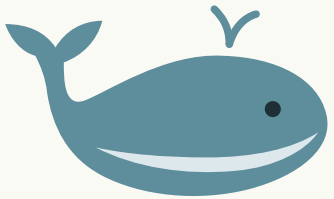
General Idea

Prove that a few bits of $f(x)$

are correct.



Our PoS Construction



x, h



Merkle root $h^*(y) = r$

succinct proof for

"$r = h^*(f(x))$"

✓ explicit det. poly-time statement

$y = f(x)$	r	π

positions: $i_1 = 3$ ... $i_\ell = 8$

bits: $y_3 = 1$ ... $y_8 = 0$

(r, π) + Merkle openings

Checks:

1. Valid Merkle openings
2. Valid proof π

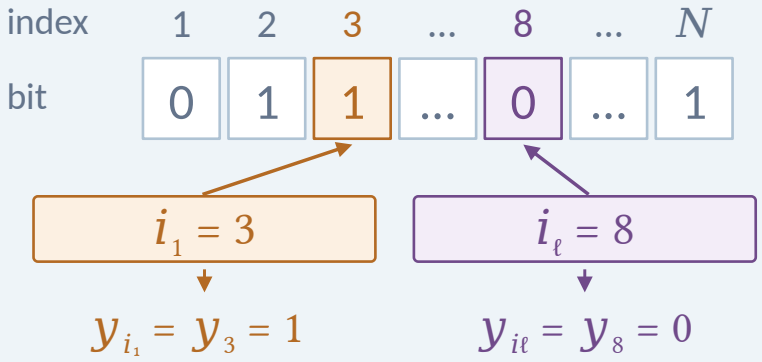
Framework

① "deep" (incompressible) f

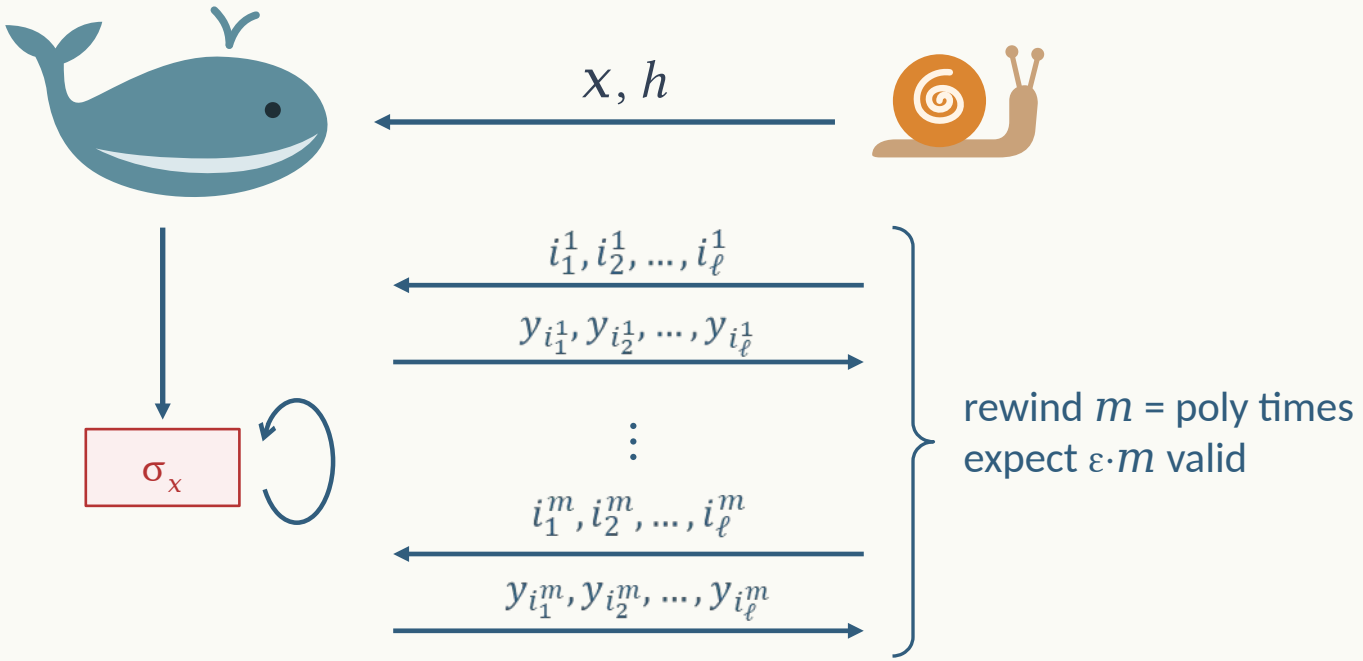
② succinct proof: storing $f(x)$

– SNARG for P

– Kilian's argument (store the PCP)



Soundness: Intuition for constant ε



Cheater: small state σ_x ($\ll N$ bits) yet passes

Rewind: proof system + Merkle tree forces answers to be true bits of $f(x)$

Extract: recover $f(x)$ from σ_x quickly

Contradiction: short, fast description of $f(x)$

Negligible soundness: encode "useful" challenges via a hitting-set generator.

Instantiating the Framework

Assume: E is hard for exponential-size nondeterministic circuits

① via SNARGs for P

also assuming SNARGs for P (+ OWF):

Proof of Space secure against $(1-o(1))N$ storage provers

nearly optimal space gap

② via Kilian's argument

also assuming collision-resistant hashing:

Proof of Space secure against $N^{1/c}$ storage provers, for some constant c

Open Problems

1

Remove crypto assumptions

Unconditional proof systems, e.g. Goldwasser-Kalai-Rothblum? Requires interaction proportional to **depth** of computation. To get $O(1)$ rounds $\longrightarrow$ f needs a low-depth circuit.

2

Non-amortizability

We want k challenges $\longrightarrow$ $k \cdot N$ space. $(G(x_1), \ldots, G(x_k)) \approx (U_1, \ldots, U_k)$ gives limited non-amortizability, but what about $k = n^c$, arbitrarily large c?

3

Memory-hard functions?

$f : \{0,1\}^n \longrightarrow \{0,1\}^n$ requiring large memory to evaluate efficiently

Thank you!

https://eprint.iacr.org/2026/1470